

Contract pricing model: technical design

Data model, contract selection, resolver and price calculation · for the development team

Status: Draft v4 proposal for Greg Smith's review. Supersedes v3 where they differ (section 14 lists every change).

Based on: Greg Smith's *Contract pricing model: design summary* (v3) and *Identification of the Billable Party* notes, revised after the 5 October 2026 review with Donald Fraser

Companion document: *How procedures are priced and who pays* gives the same rules in business language for AA directors, anaesthetists and the office

1. Purpose

This model holds the pricing inputs for anaesthetic procedures and decides who pays. It rests on one separation: the **procedure** records what the anaesthetist did, and the **contract** records how it is paid for. They are kept apart because the same procedure is paid in different ways for different surgeons, hospitals, insurers and anaesthetists.

The RVG is an anatomically grouped list of recommended base units. The design challenge is to support the many specific procedures within those groups, overlaid with contractual pricing, without creating a blizzard of combinations. The model meets it with three ideas:

- **Defaults live once.** RVG groups hold the published units. Procedures inherit them and override only where needed.
- **Contracts are sparse overrides.** A contract line carries only the fields it sets. Everything else shows through from below.
- **One path for every booking procedure.** "No contract" is modelled as a default contract, so the resolver and the billing engine never branch on whether a contract exists.

This is a logical model. Data types, keys and indexes are left to the physical design.

2. Terms

Term	Meaning	Business wording
Booking	One patient's appointment on a List. Holds one or more booking procedures.	The booking; anaesthetists say "the card"
Procedure	An entry in the master procedure list. A child of one RVG group.	Procedure
Booking procedure	One procedure performed on one booking, with its selected contract, recorded units and price. Replaces v3's "episode".	Procedure
Time entry	The anaesthetist recording BTM units, modifier notes, and any price or discount against a booking procedure in the mobile app. Replaces v3's "timesheet".	Filling in the card
Contract holder	The party a contract belongs to: an insurer, hospital, surgeon's rooms, or an anaesthetist (for their own price list).	Contract holder
Contract	A dated set of pricing terms belonging to one holder. Not necessarily a legal contract.	Contract
Contract line	The terms for one procedure, or one whole RVG group, within a contract.	A line on the price list

Term	Meaning	Business wording
Default contract	The single contract flagged <code>is_default</code> . It has no holder and no lines.	“No contract (RVG)”
Third party / first party	A third-party contract is an agreement with someone else, and its prices are locked. A first-party contract is the anaesthetist’s own pricing, which they can adjust.	Same
Billable party	Whoever receives the invoice: the holder’s billable party record, or the payer named on the booking.	Who pays
Payer	The name and email of the person paying, captured on the booking. Prefilled from the patient and editable (for example to a guardian).	Patient or guardian
BTM	Base, time and modifier units.	BTM
Resolver	The function that returns the starting units and contract terms for a procedure and contract as at a date.	Not used
Price source	Which rule produced the final price (section 8).	Not used

3. Data model

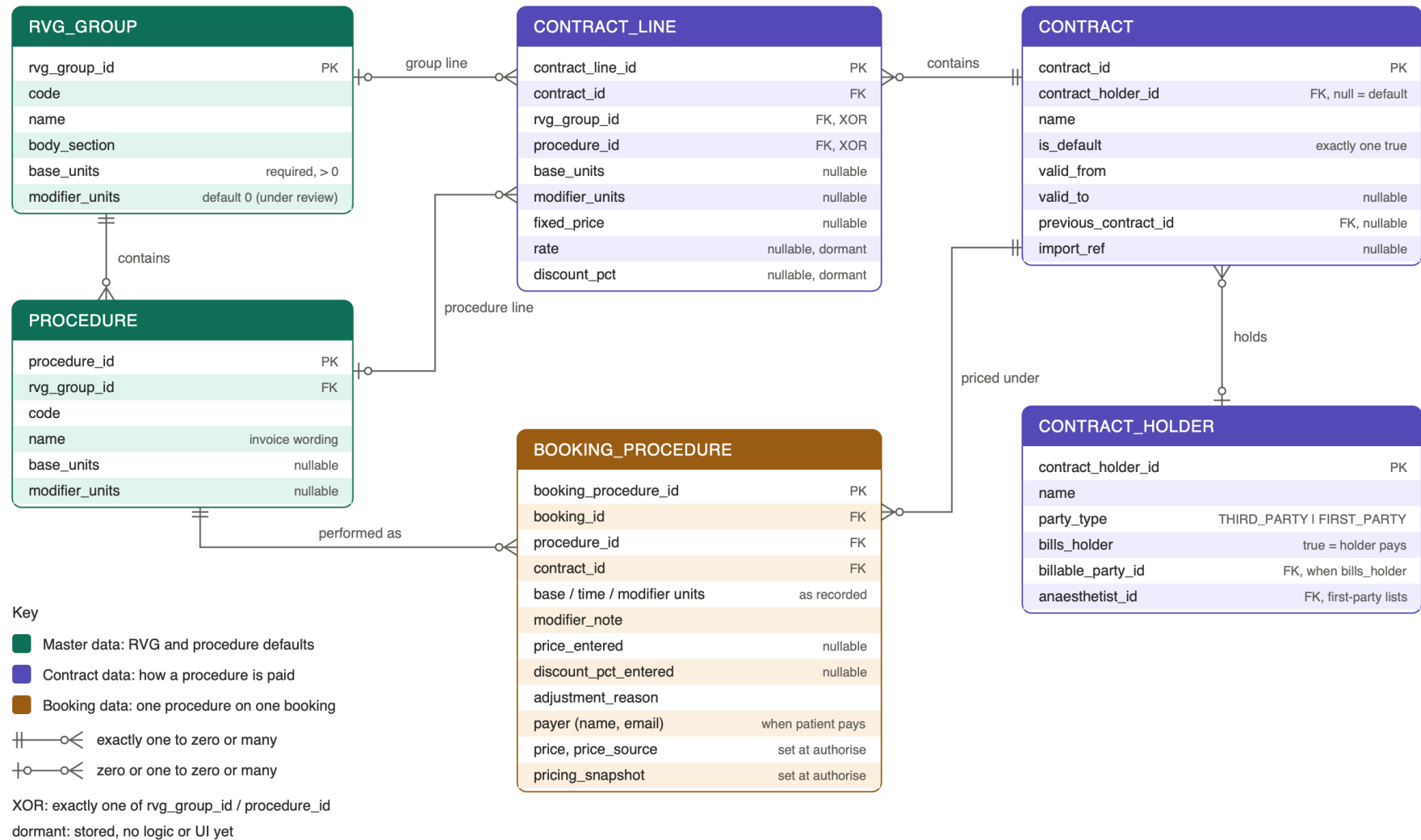


Figure 1. Revised contract pricing model. Precedence rises from the master data (green) to the contract overrides (purple). BOOKING_PROCEDURE shows only the fields pricing needs.

RVG_GROUP (master data)

The published RVG group, as listed in the NZSA guide. AA may add its own groups.

Field	Rule	Notes
rvg_group_id	PK	Surrogate key. RVG codes are not unique enough to be keys.
code, name	Required	As published.
body_section	Required	Top level of the picker (for example Head). Could be its own table.
base_units	Required, > 0	The bottom layer of the resolver, so base units always resolve.
modifier_units	Required, default 0	Stored as an explicit 0 where the RVG has none. Under review (section 15).

PROCEDURE (master data)

The specific procedure, worded as it should appear on the invoice. A procedure need only be defined when AA needs it.

Field	Rule	Notes
procedure_id	PK	
rvg_group_id	FK, required	Exactly one parent group. No nesting of procedures.
code, name	Required	name is the invoice wording.
base_units, modifier_units	Nullable	Null inherits from the group. Typically null.

CONTRACT_HOLDER

The party a contract belongs to. Two new behaviour fields replace v3's assumption that the holder is always billed.

Field	Rule	Notes
contract_holder_id	PK	
name	Required	For example "Southern Cross", "Merivale Plastics", "Dr B. Smith prepaid list".
party_type	THIRD_PARTY or FIRST_PARTY	Third party: an insurer, hospital or surgeon's rooms; prices locked. First party: an anaesthetist's own pricing; adjustable.
bills_holder	Boolean	True when the holder pays AA (insurer, hospital). False when the holder only sets the price and the payer on the booking is billed (surgeon fixed-fee arrangements, anaesthetist lists).
billable_party_id	FK, required when bills_holder	Links to the existing billable party record.
anaesthetist_id	FK, nullable	Set for first-party lists. Their contracts are offered only on that anaesthetist's bookings.

CONTRACT

A dated set of terms. A holder can have many contracts. Each price review creates a new contract row (section 11).

Field	Rule	Notes
contract_id	PK	
contract_holder_id	FK, null only for the default contract	
name	Required	Display name. The default contract shows as “No contract (RVG)”.
is_default	Exactly one true	The default contract is found by this flag, never by its name.
valid_from, valid_to	valid_to nullable	Validity sits here, not on lines (changed from v3). Null valid_to means open-ended.
previous_contract_id	FK, nullable	Links a price-review version to the contract it replaces.
import_ref	Nullable	Points to the import log entry that created this version.

CONTRACT_LINE

A contract’s terms for exactly one procedure or exactly one RVG group. Every pricing field is nullable, so a line carries only what it sets.

Field	Rule	Notes
contract_line_id	PK	
contract_id	FK, required	
procedure_id / rvg_group_id	FK, exactly one set (XOR)	A group line covers every procedure in the group that has no procedure line of its own. None are known today.
base_units, modifier_units	Nullable	Override the units beneath. 0 is a value, not “nothing”.
fixed_price	Nullable	The whole price for the procedure. BTM becomes informational.
rate	Nullable, dormant	A contract unit rate in place of the anaesthetist’s. Not used by AA today.
discount_pct	Nullable, dormant	A contract-level percentage discount. Not used by AA today.

BOOKING_PROCEDURE (pricing fields only)

Field	Set by	Notes
procedure_id, contract_id	Office at setup; anaesthetist may change	One contract per booking procedure.
base_units, time_units, modifier_units	Anaesthetist at time entry	Base and modifier start from the resolver. Time is entered.
modifier_note	Anaesthetist	Required when modifier units are claimed. Insurers ask for it.

Field	Set by	Notes
price_entered	Anaesthetist, adjustable contracts only	Stored only when the anaesthetist types a price. Null otherwise.
discount_pct_entered	Anaesthetist, adjustable contracts only	Percentage only. A flat change is made through price_entered.
adjustment_reason	Anaesthetist	Required with a price or discount.
payer_name, payer_email	Office or anaesthetist	Used when the payer is billed. Prefilled from the patient.
price, price_source, pricing_snapshot	Billing engine at authorisation	Section 8.

4. Contract behaviour

Two holder fields drive everything the anaesthetist sees and who is billed. One derived rule decides whether the anaesthetist may adjust the price:

Adjustable rule

```
adjustable = contract.is_default OR holder.party_type = FIRST_PARTY
```

Contract	Billed	Price field at time entry	Discount %
Default ("No contract (RVG)")	Payer on booking	Empty and editable (a nominated price)	Allowed
First party, for example an anaesthetist's prepaid list	Payer on booking	Shows the resolved fixed price, editable	Allowed
Third party, bills_holder true (insurer, hospital)	Holder's billable party	Shows the resolved fixed price, read-only	Hidden
Third party, bills_holder false (surgeon fixed-fee arrangement)	Payer on booking	Shows the resolved fixed price, read-only	Hidden

The office can override any price at review, with a reason. That override is the only way to change a third-party price without changing the contract.

5. Contract selection

1. **Office, at booking setup.** The office picks the procedure (body section, then group, then procedure), then selects or confirms the contract.
2. **Candidate list.** The default contract always appears, first. It is followed by every contract that is valid on the procedure date, has a line for the procedure or its group, and whose holder fits the booking's context: the List's hospital, the surgeon or rooms, the insurer, and for first-party lists the booking's anaesthetist. The exact context rules are open (section 15).
3. **Anaesthetist, at time entry.** The anaesthetist may change the procedure or the contract until the list is submitted. No reason is needed. The change is audited and flagged at office review.
4. **On any change,** the resolver re-runs and the starting units and fixed price refresh. If the new contract is not adjustable, any price_entered and discount_pct_entered are cleared.

Fixing the default-contract navigation flaw in v3

In v3 the default contract has no lines, so a lookup from procedure to contract lines never finds it. Here the default contract is offered for every procedure **by rule**, not by line lookup. The alternative is

to auto-create an empty default line for every procedure. That behaves identically but adds one empty row per procedure. Developers may choose either.

6. The resolver

Think of the layers as a stack of transparent sheets. For each field, look down through the stack, and the first sheet with a value wins.

Inputs

A procedure, the selected contract (which may be the default) and the **procedure date**. Only a contract valid on that date is used, whenever the resolver runs.

Layers, highest precedence first

1. The selected contract's line for the procedure
2. The selected contract's line for the procedure's RVG group
3. The procedure (base_units, modifier_units)
4. The RVG group (base_units, modifier_units)

base_units and modifier_units resolve through all four layers. Because the RVG group is fully populated, they always resolve. fixed_price, rate and discount_pct exist only on contract lines, so they resolve from layers 1 and 2 or stay null.

v3 also stacked the default contract's lines beneath the selected contract. Those layers are dropped because the default contract has no lines (section 12). A practice-wide price on the default contract would amount to AA setting a common price for its members, which competition law does not allow.

Output

The resolved values, plus the layer each one came from. Both go into the pricing snapshot.

Treatment of nulls

- Null means "not specified here", and resolution looks through to the next layer. A value, including 0, stops it.
- An explicit 0 above the RVG level is only needed to switch off a non-zero value beneath. That is rare, but some contracts do zero modifiers, so the distinction matters.
- In editing screens, each overridable field shows the inherited value as a greyed placeholder, with an explicit choice between **inherit** (stored as null) and **override** (an override with no value stores 0).
- In import spreadsheets, a blank cell means inherit and an explicit 0 means zero.

7. Time entry (the card)

- The resolved base and modifier units are shown as starting values. The anaesthetist enters time units and may change base and modifier units. Out-of-range values warn the office. They are never blocked.
- Modifiers are itemised, with a required modifier_note when any are claimed. Whether modifiers are ever pre-filled is open (section 15).
- The price field and the discount follow the contract behaviour table in section 4.
- The record keeps the values the anaesthetist recorded. Starting values can always be recomputed by the resolver, and they are captured in the snapshot at authorisation.

8. Price calculation

The billing engine prices each booking procedure on its own when the list is authorised.

Validation first

The engine rejects the booking, rather than ignoring one input, when:

- `price_entered` or `discount_pct_entered` is present on a contract that is not adjustable
- BTM units are missing and no fixed price resolves
- the selected contract is not valid on the procedure date

Precedence

Step	Condition	Price	price_source
0	The office overrode the price at review	The office price	OFFICE_OVERRIDE
1	<code>price_entered</code> is present (adjustable contracts only)	<code>price_entered</code>	ANAESTHETIST_PRICE
2	<code>fixed_price</code> resolves	<code>fixed_price</code> . Recorded BTM is informational only.	CONTRACT_FIXED_PRICE
3	Otherwise	$(B + T + M) \times \text{rate} \times (1 - \text{discount_pct_entered} / 100)$, where rate is the resolved contract rate if any (dormant), else the anaesthetist's own unit value	CALCULATED

- A price of 0 produces a no-charge invoice. BTM is still recorded.
- Whether the RVG multi-procedure rule (for example base units on the primary procedure only) still applies is being confirmed with AA. If it does, it adjusts the units of CALCULATED procedures before step 3.

Snapshot

At authorisation, each booking procedure stores the procedure, the contract (each version is its own row, so the id is enough), the resolved values and their layers, the recorded BTM, the rate and discount used, the price, the `price_source`, and the billable party or payer details. Later changes to the RVG, procedures or contracts never alter historical billing, and anaesthetist-nominated prices are easy to report on.

9. Billable party

Selected contract	Invoice goes to
Default contract	Payer on the booking (patient by default, editable to a guardian)
Holder with <code>bills_holder</code> true	The holder's billable party record
Holder with <code>bills_holder</code> false	Payer on the booking

How this maps to the billable-party truth table:

Patient insured (insurer pays AA)	Contract applies	Outcome	Configuration
Yes	Yes	Contract holder billed	Third-party holder, <code>bills_holder</code> true

Patient insured (insurer pays AA)	Contract applies	Outcome	Configuration
No	Yes	Patient billed at the contract price	Third-party holder, <code>bills_holder</code> false (for example a surgeon's fixed-fee arrangement)
No	No	Patient billed at RVG	Default contract
Yes	No	Inconsistent	Warning at office review, not a block (see below)

- “Insured” here means an insurer pays AA directly. Insured patients who claim the money back themselves are on the default contract.
- The surgeon’s booking or the hospital may indicate insurance status (for example SXAP). If the booking carries such an indication, a payer-billed contract with an “insurer pays” indication raises a warning at review. It is not a block, because pre-approval regularly falls through on the day and the anaesthetist then switches to the default contract and bills the patient. Where this indication is stored is open.
- AA acts as agent for independent anaesthetists, so the system accepts the anaesthetist’s choice of contract and relies on office review to catch mistakes.

10. Prepayment

Prepayment uses the pricing model but is specified elsewhere. The points that touch this model:

- **Trigger.** The booking procedure’s procedure, or its RVG group, is in the anaesthetist’s prepaid set, and the payer is a person, never an organisation.
- **Amount: open.** Option (a) models it as a first-party contract per anaesthetist (`party_type` `FIRST_PARTY`, `anaesthetist_id` set, `bills_holder` false, lines with `fixed_price`), so the prepaid amount is simply the resolved fixed price and needs no new structures. Option (b) estimates it from expected units at the anaesthetist’s own unit value. The current requirements specify (b). The 5 October discussion leaned towards (a), because each anaesthetist must set their own prices. The directors are to decide.
- **Settlement (proposed, to confirm).** The prepaid amount behaves as an adjustable fixed price. With no `price_entered`, the final price equals the prepaid amount and nothing more is billed. If the anaesthetist raises the price, the difference is billed as an additional invoice. If the final price is lower, nothing is refunded. A cancellation refunds in full from trust. The current requirements instead invoice any positive BTM balance automatically, so this needs a decision.
- The link from a booking procedure to its prepayment belongs on the booking procedure.

11. Updates, history and data loading

Contract versions

- Validity is held on `CONTRACT` only. A price review creates a new contract row with `previous_contract_id` set and a new `valid_from`. Its lines are copied from the previous version and then changed, and the previous version’s `valid_to` is set to the day before.
- A **delta** upload (the usual case) copies all lines and applies only the rows supplied. A **full** upload creates the version from the supplied rows alone, so any line not present is closed with the old version rather than deleted.
- A row marked `REMOVE` is not carried into the new version, so that target falls back to the procedure and RVG layers.
- A row is the complete line, not a patch: a blank cell inherits and 0 means zero.

- A preview lists new, changed and removed lines for approval before anything is applied. `import_ref` points to the import log.

Data loading

- All master data reaches AA as spreadsheets in AA-defined templates: RVG groups, procedures, contract holders, contracts and lines, plus hospitals, surgeons and rooms, and anaesthetists.
- Loading must be repeatable, so test environments can be wiped and reloaded.
- The loading mechanism is open. The suggested MVP baseline is a developer-run import, then office screens to create, edit and retire records. An in-app upload with preview, and exporting a contract to the template for its holder to update, are later options.

History

Creates, updates and retirements of RVG groups, procedures, contract holders and contracts are audited (who, what, when). That replaces v3's suggestion of modelling the RVG as the lowest contract to keep its history. Billing history is covered by the snapshot.

12. Integrity rules

- `CONTRACT_LINE`: exactly one of `procedure_id` and `rvg_group_id` is set.
- At most one line per contract per target.
- No overlapping validity between a contract and its previous or next version.
- Exactly one contract has `is_default` true. Its `contract_holder_id` is null, it has no lines, and it never expires.
- `RVG_GROUP`: `base_units` is required and greater than 0, and `modifier_units` is required with a default of 0.
- `bills_holder` true requires `billable_party_id`.
- A holder with `anaesthetist_id` set is `FIRST_PARTY`.
- The billing engine rejects a price or discount on a contract that is not adjustable.

13. Dormant capabilities

These are in the data model but have no logic or screens yet. AA uses only RVG pricing and fixed prices today. Keeping the columns means enabling them later needs a software release, not a schema change.

- `CONTRACT_LINE.rate`: a contract unit rate
- `CONTRACT_LINE.discount_pct`: a contract-level percentage discount
- RVG group lines: one price for every procedure in a group. None are known today, but the XOR structure supports them.

14. Changes from v3

v3	v4	Why
"Episode", "timesheet", "card"	Booking procedure, time entry	One vocabulary across the documents and requirements.
<code>valid_from/valid_to</code> on contract lines	Validity on <code>CONTRACT</code> only. Price reviews create versions.	All lines in a contract share its dates (Greg, 5 Oct).
Default contract reached through its lines	Offered for every procedure by rule. Resolver has four layers.	The default contract has no lines, so the v3 navigation could not find it.

v3	v4	Why
Holder always billed via billable_party_id	bills_holder flag and party_type. Payer captured on the booking.	Surgeon fixed-fee arrangements set the price but the patient pays.
"Timesheet price overrides all other logic"	Only on adjustable contracts. Third-party fixed prices are locked.	Third-party agreed prices cannot be changed by the anaesthetist.
Timesheet price valid only with no contract	Also valid on first-party contracts	Anaesthetists keep discretion over their own pricing, including prepaid amounts.
Prepaid difference becomes a balance due or a refund	An increase is invoiced. No refund (to confirm).	Agreed in the 5 Oct review. AA does not refund prepayments.
"Fixed base units" contract parameter	"Base units" override	They are starting values, not fixed.
Practice-wide self-pay price via default contract lines	Removed	AA cannot set a common price for its members.
Billing issues a receipt, not an invoice, when prepaid	Removed	Not confirmed. Handled by the prepayment specification.
modifiers_excluded flag; per-modifier child table	Removed	Out of scope. The anaesthetist itemises modifiers in the note.
RVG history via a lowest-precedence contract	Audit records on update	Simpler, and sufficient.
Full upload removes missing lines	Closes them with the old version	Nothing is deleted.
Override units "with an explanation"	Note required for modifiers and for price adjustments. None for a contract change.	Insurers ask for modifier explanations. Contract changes are flagged at review instead.

15. Open points

To be decided

Modifiers. Ben's view is that there are no modifier units in the schedule and nothing should be auto-applied. The current requirements pre-fill modifiers from ASA grade and from procedure defaults. The directors discuss this on 6 October. If Ben's view holds, `modifier_units` stays at 0 throughout and the RVG page 12 guidance may be offered as a filtered pick-list.

Prepayment amount and settlement: options (a) and (b) in section 10, and whether the anaesthetist sees the prepaid amount at time entry.

Where `adjustable` lives. Derived here from the holder. Move it to `CONTRACT` or `CONTRACT_LINE` only if one holder needs mixed behaviour.

Insurance indication on the booking: whether and where to store it for the section 9 warning.

Candidate list context rules: hospital, surgeon or rooms, insurer, anaesthetist.

RVG multi-procedure rule: whether it still applies.

Data loading mechanism for the MVP.

Default contract offering: by rule, or empty lines per procedure.

Out of scope for this document

The anaesthetist's unit value (held on the anaesthetist profile), the booking and list lifecycle, invoice generation, the import log, and the billable party record are referenced but not modelled here. No nesting or versioning of individual lines or procedures is provided. Each stands alone.